

OmniTrax®

Ranging Buried Ported Coaxial Sensor

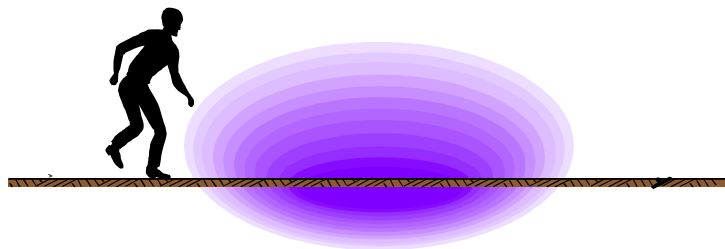
Guided Radar

- General Properties
 - Uniform electromagnetic field
 - Terrain-following, covert
 - Well contained volumetric detection
 - Optimum detection of humans
 - Maximum rejection of small animals
 - Immune to vegetation and blowing debris

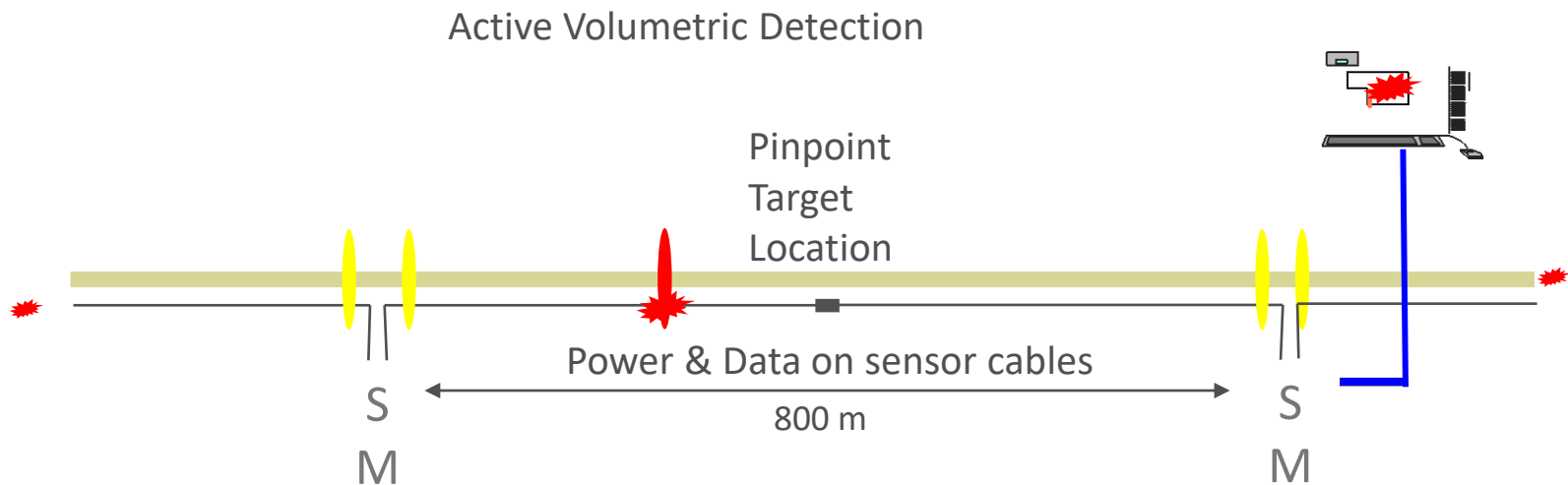


Guided Radar

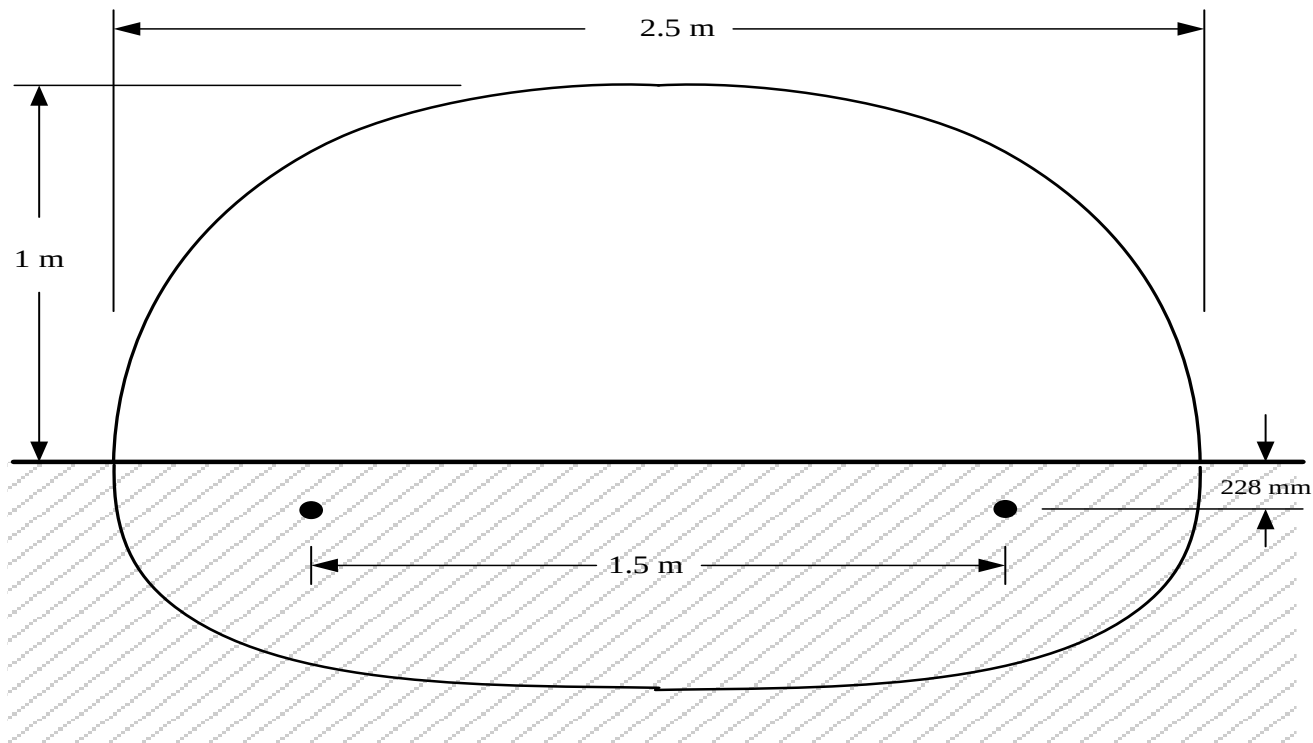
- Coded Pulsed
 - Time Domain Reflectometry (TDR) location of targets to within 1 m (3.3 ft) using time delay
 - Up to 50 zones per processor



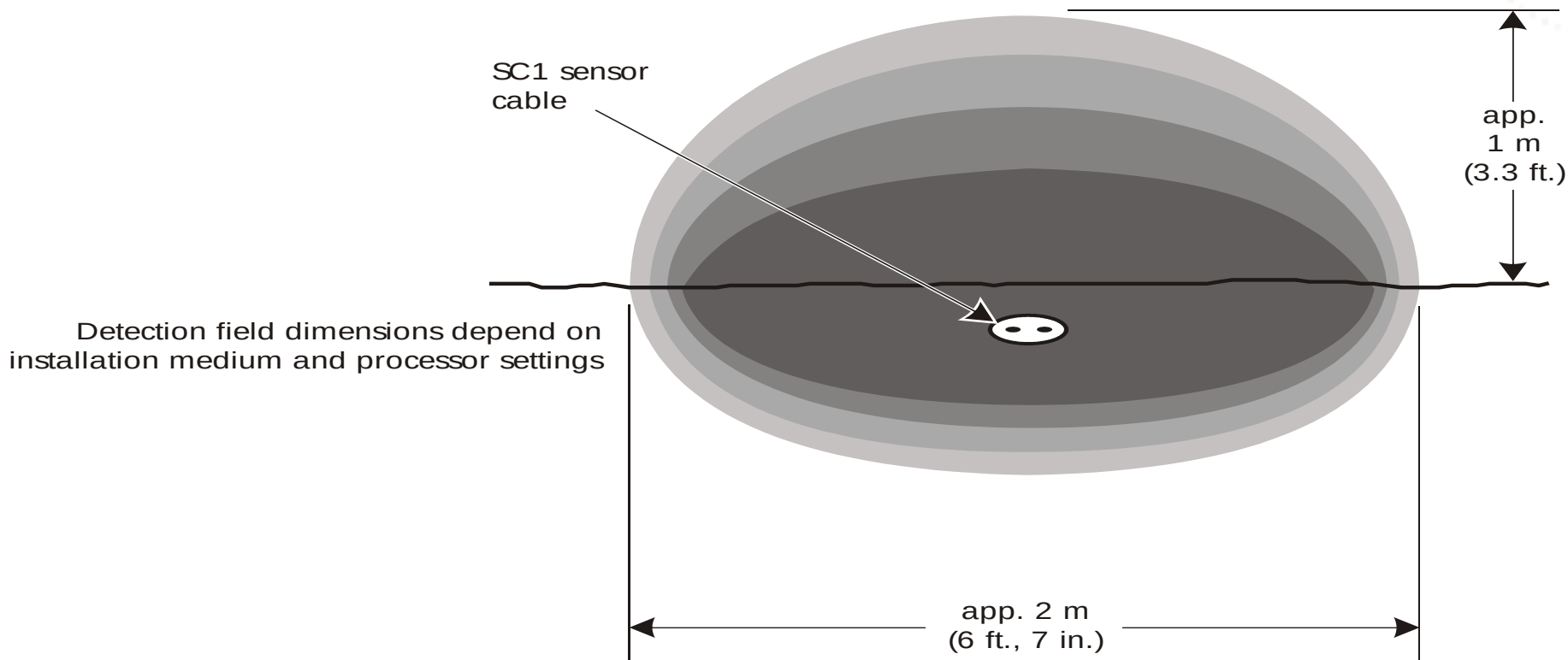
Ranging



Uniform Detection Zone – Cross Section



SC1 Detection Zone – Cross Section



What is Ranging Technology?

- OmniTrax transmits pulse-coded ultra-wideband RF signals between 27 MHz and 37 MHz, close to the 40 MHz used in Perimitrax
 - OmniTrax still operates in the frequency band that is ideal for human detection
- Return signals are received and processed as follows
 - **Location** – signals of interest are identified and located to the nearest meter (3.3 ft)
 - **Qualification** – the detection peak signal is compared to the combined threshold for that meter
 - **Annunciation** – if the signal is above threshold, then alarm information is used to trigger a local relay and / or is sent on the Silver Network through the Network Interface Unit (NIU) and the Silver Network Manager to StarNet

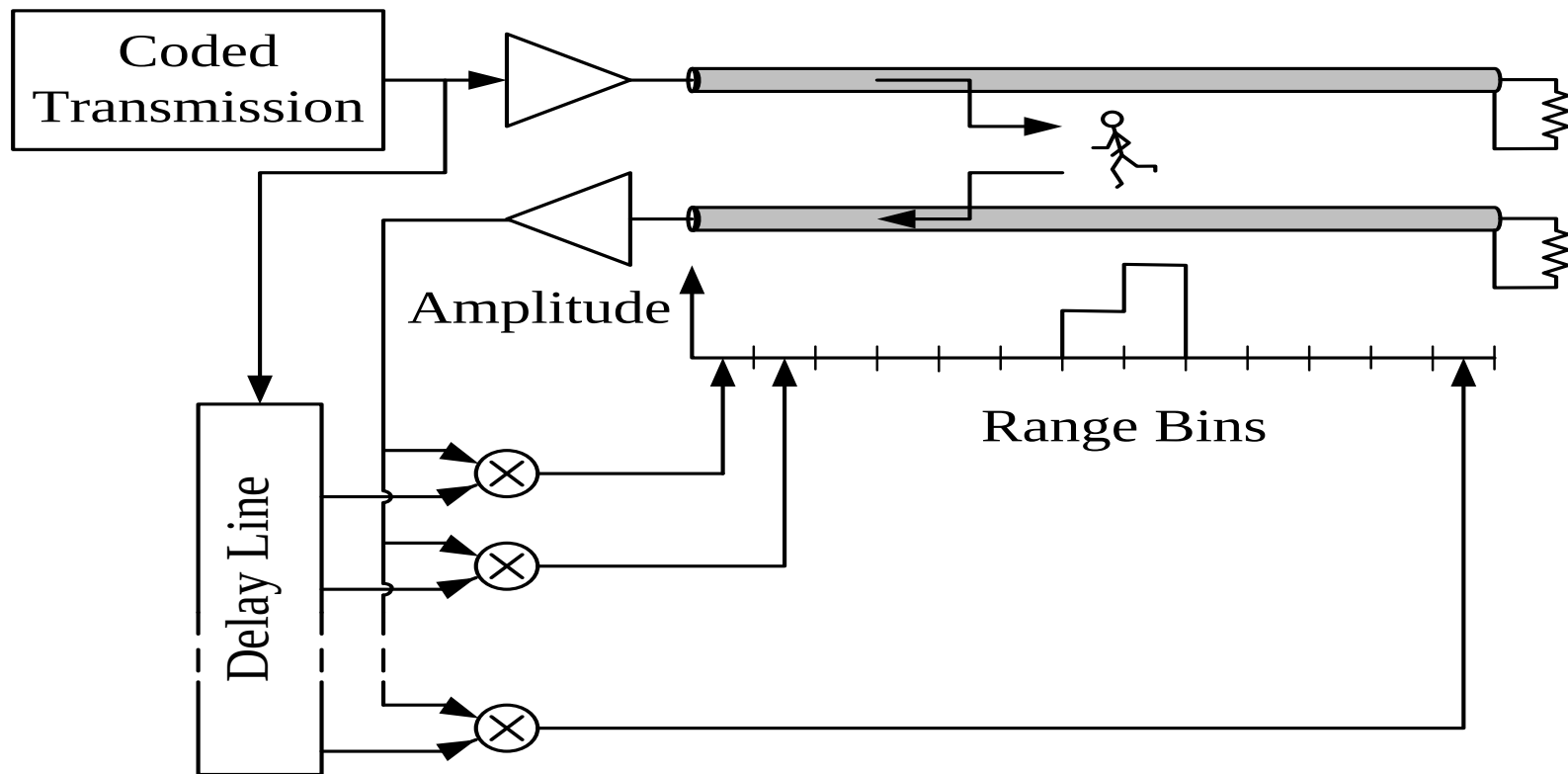
Ranging Process

- Transmitted and received pulses are compared in a “Screaming Correlator” to identify the general location of a “pulse of interest”



- Further processing and pre-qualification locates the peak of the signal to the nearest meter (3.3 ft)
- The signal is compared to the combined threshold for that meter
 - If it is below threshold, it is deleted / rejected
 - If it is above threshold, an alarm is declared

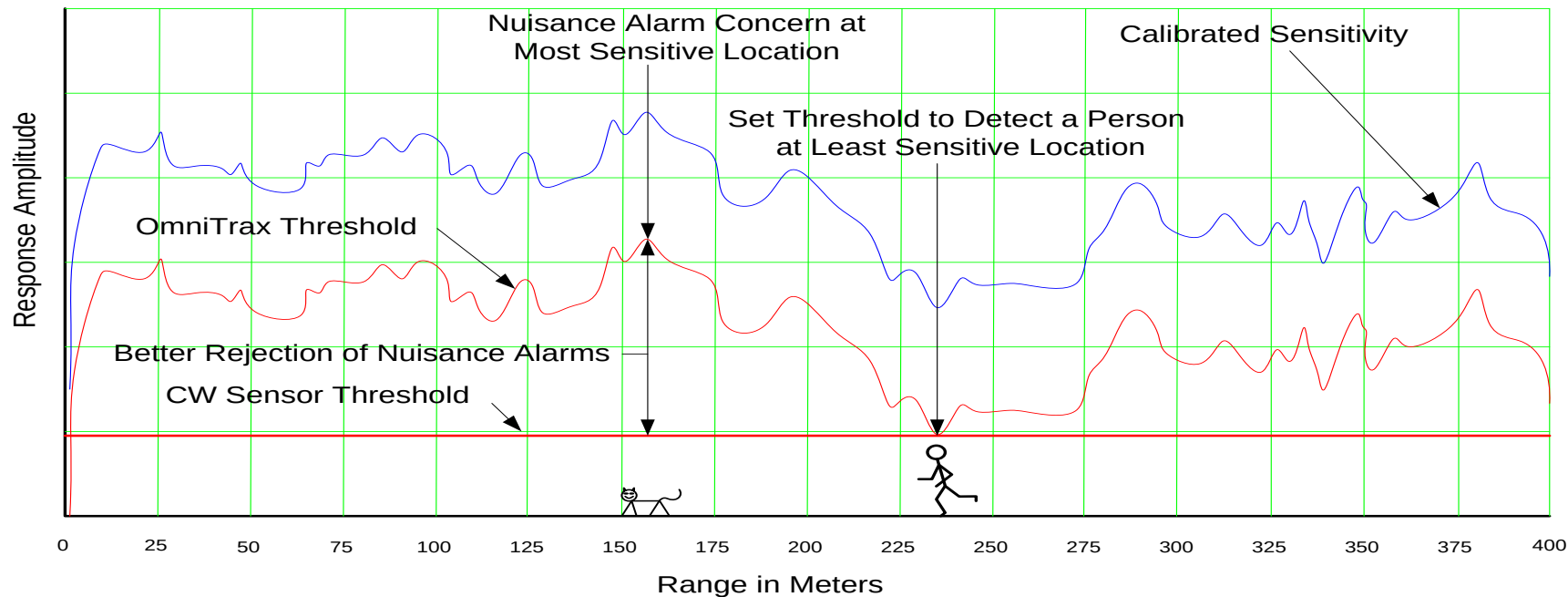
Ranging Technology



Thresholding

- During commissioning, while OmniTrax is in calibrate mode, a person walks between the cables or on the cable
 - OmniTrax records a sensitivity profile for every meter (3.3 ft) of cable
- A global margin is applied to the sensitivity profile to set a specific threshold for each meter
 - In absolute terms, every meter will have a different threshold
 - In relative terms, every meter will have the same sensitivity
 - The default global margin can be changed

Calibrated Thresholding



Thresholding

- Further thresholding can be applied to increase or decrease the sensitivity of a particular portion of the perimeter
 - Software segments (up to 50 per cable) are created for each part of interest
 - All meters in that segment will have the same new sensitivity
 - The segment threshold is used to make a section of the cable non-reporting

Thresholding

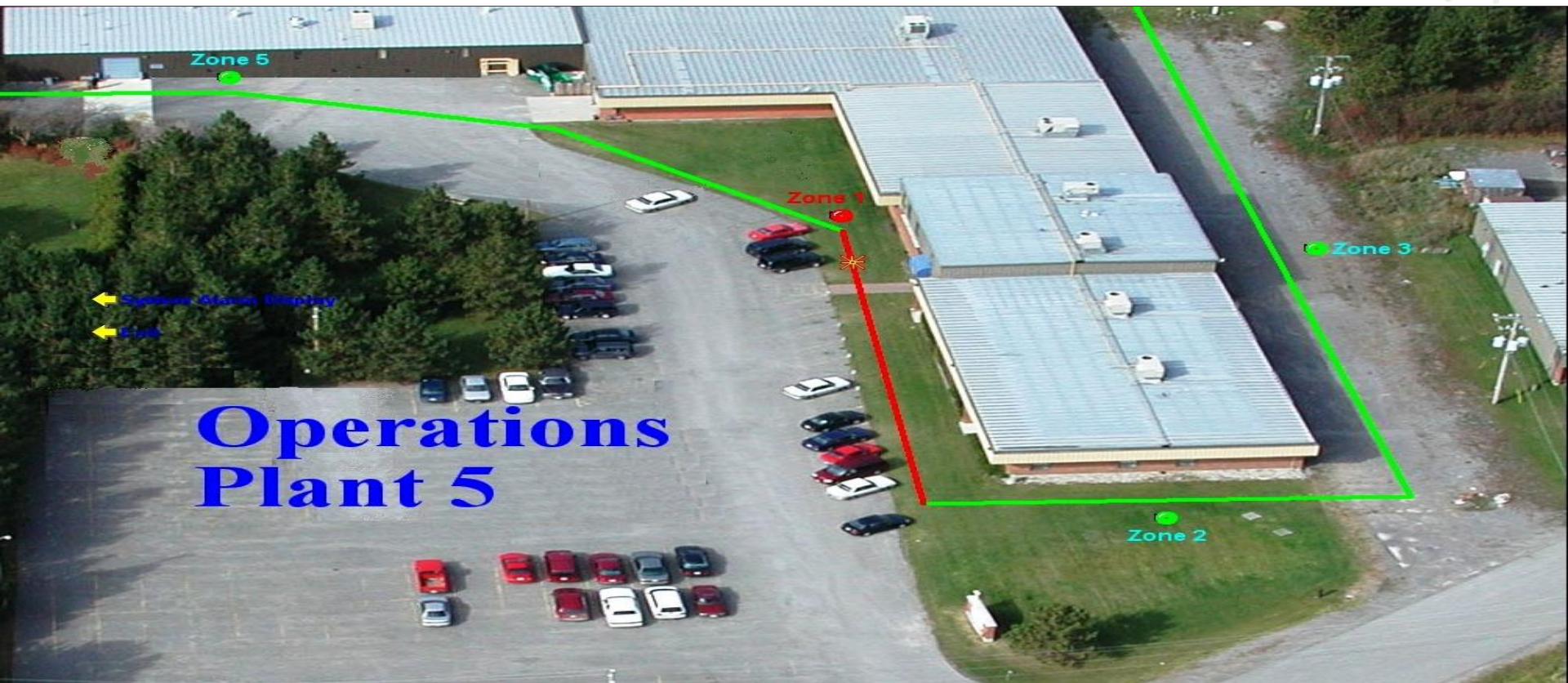
- Software segments are assigned to a detection (or display) zone
 - A detection zone is equal to one or more segments
 - Segments making up a zone do not need to be contiguous and can bridge between the A and B cables
 - A detection zone must start and end at the start and end of a software segment (not necessarily the same software segment)
- The location of the start and end of each alarm zone is programmed into StarNet in absolute meters
 - This information is obtained from the UCM configuring the processor

Thresholding

- If a zone on a map is not a single straight line, display lines have to be drawn on the map to mimic the actual perimeter
- The start and end of each display line of the perimeter is entered into StarNet in absolute meters
- On alarm, intrusion location is supplied to StarNet by the Silver Network Manager
 - The location is specified by the processor number, the zone number and the exact location to the nearest meter

Thresholding

- StarNet annunciates the alarm by
 - Turning the corresponding display line red
 - By dynamically placing, on that display line, an icon to show the exact location of intrusion to the nearest meter
- The choice of changing the state of a display line on the map, rather than the entire zone, is done to make it easier to locate the intruder with reference to geographical landmarks on the perimeter (e.g. 5 m or 916 ft) from the single gate on the east side)



Performance Benefits of Location

- Calibrated Thresholding
 - Uniform detection zone cross section
 - Reduced Nuisance Alarm Rate (NAR)
 - Relaxed installation criteria (lower cost)
- Software Segmentation
 - Create inactive cable (cross driveways – no splicing)
 - Performance customization

Performance Benefits of Location

- Precise Diagnostics
 - Real-time display of processor performance
 - Display of archived data
 - Flash memory recordings (replaces strip chart recorder)
 - Pinpoint the location of problems (loose fence, water erosion, etc.)

Ease of Installation

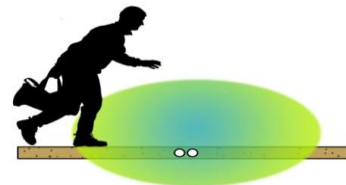
- Fewer Connectors
 - Continuous lead-in and lead-out
- Fewer Processors
 - Multiple zones per cable
- Rugged Cable
 - Direct burial other than in extreme conditions

Ease of Installation

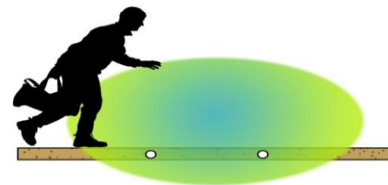
- Mixed Medium Capability
 - Sod, asphalt, concrete, etc.
- Relaxed Installation Tolerances (depth and width)
 - Reduced cost of conventional trenching
 - Potential use of a cable plough

“Short” Cables

- SC1 – Tx and Rx cables are in one cable jacket up to 200 m (656 ft) long
 - Single trench for standard detection
 - Integrated lead-in cable
- SC2 – Tx and Rx cables are two separate cables up to 200 m (656 ft) long
 - Two trenches for wider detection field in open areas
 - Integrated lead-in cable



SC1



SC2

“Short” Cables

- Standard lengths for SC1 and SC2
 - 50, 100 and 200 m (164, 492 and 656 ft)
 - Cut cable to the desired cable length to fit site requirements
- High density polyethylene jacket with flooding compounds: rugged, abrasion and chemical resistant

“Short” Cables

- SC1

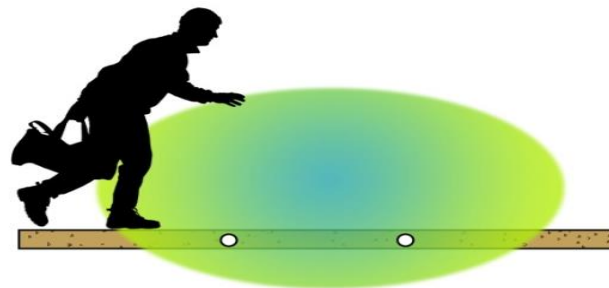
- SC2

- OC2



New Sensor Cables

- OC2 – Tx and Rx cables are two separate cables up to 400 m (1,312 ft) long
- Standard lengths for OC2
 - 300 m (984 ft)
 - 400 m (1,312 ft)
- High density polyethylene jacket with flooding compounds: rugged, abrasion and chemical resistant



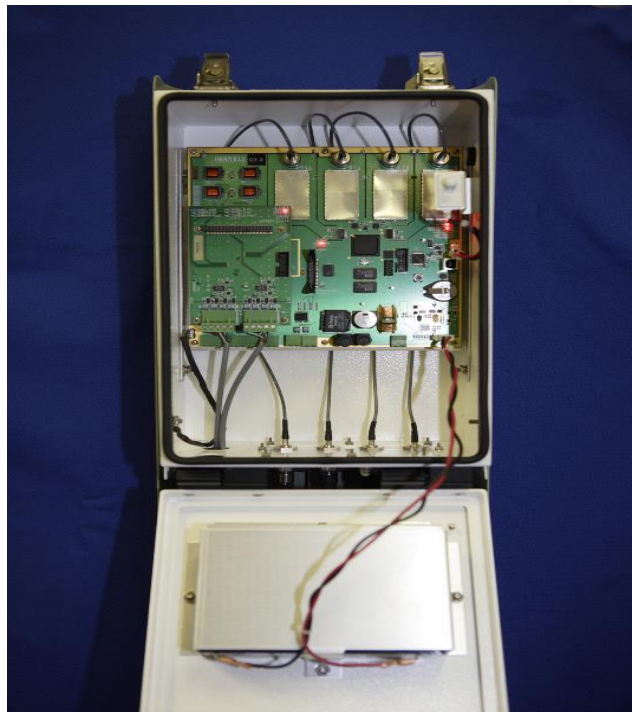
Sensor Processor Installation

- The OmniTrax processor fits in the enclosure
- The old Perimitrax telecom enclosure as shown
- OmniTrax has its own wider telecom

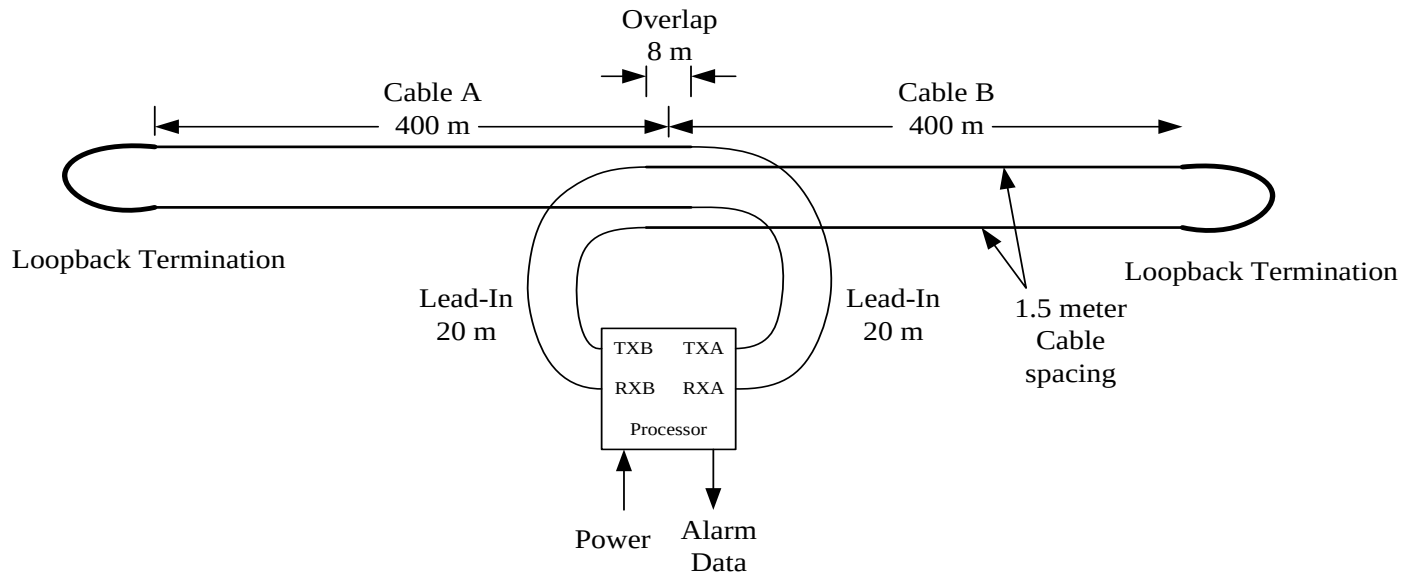


Sensor Processor

- Provides Tx, Rx and advanced microprocessor for signal processing of two sensor cables
- Uses UWB coded pulse technology (patent pending)
- New calibrated thresholding per meter (3.3 ft) enhances performance under extreme site and environmental conditions
- Determines the location of the intruder to within 1 m (3.3 ft)
- Designed for -40° to 70° C (-40° to 158° F)



Standard Configuration



Single Processor Configuration

- Excludes
 - Data communication on the cables
 - Power distribution on the cables
 - Ranging data output
- Includes
 - Perimeter coverage for up to 800 m (1/2 mile)
 - Form C relay outputs – Alarm A, Alarm B, supervision and fail
 - Self-test inputs A and B
 - Calibrated thresholds per meter and software segmentation
 - Full UCM diagnostic capability

Single Processor Configuration

- Options (select one not both)
 - Output card adds an additional eight Form-C relay outputs that are set up with the UCM
 - Input card adds up to eight supervision inputs that are set up with the UCM

Networking Options

- Communications Card (optional add-on card)
 - Each node regenerates (amplitude and timing) the message
 - Up to 32 nodes per network
 - Five types of cards
 - RS-422
 - Multimode fiber
 - Single mode fiber
 - RS-422 and multimode fiber
 - RS-422 and single mode fiber

Networking Options

- Data Over Sensor Cable
 - Up to 32 processors
 - Each processor must be equipped with a network add-on card
- Power Over Sensor Cable
 - 12 to 48 VDC, 9 watts per processor
 - Up to five processors for a single 48 VDC power supply

Silver Network and Network Interface Unit (NIU)

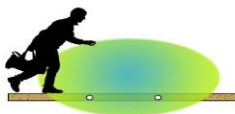
- Multimode NIU
 - Connects Network Manager running on PC to the network
 - RS-422 or multimode fiber network
 - USB / RS-232 / Ethernet (TCP/IP) to PC
- Single Mode NIU
 - Connects Network Manager running on PC to the network
 - RS-422 or multimode fiber network
 - USB / RS-232 / Ethernet (TCP/IP) to PC

Silver Network and Network Interface Unit (NIU)

- Silver Network
 - Point-to-point (RS-422 / fiber or over sensor cable)
 - Regenerative at each node
 - Up to 32 nodes
 - Supports redundant operation

Single Network Limits

- Power in groups of 5 SPs each with 2 x 400 m (1,312 ft) cables
 - Total cable length possible per group is 4 km (2.5 mi)
 - Power is supplied to the middle SP in each group
 - The limit of five SPs also applies to SC1 and SC2 cables
- Multiple networks can extend perimeter protection without limit



Single Network Limits

- Data, 32 SPs, 64 Cable Lengths
 - Possible perimeter length is $64 \times 400 \text{ m}$ (1,312 ft) = 25.6 km (15.9 mi)
 - At the point of extraction, the RS-422 copper data cable length to the NIU cannot exceed 1200 m (4,000 ft)
 - At the point of extraction, the multimode fiber optic cable length to the NIU cannot exceed 2,200 m (7,200 ft)

UCM Software

- A Windows®-Based Software Utility
 - Laptop or desktop PC
 - USB connection to signal processor or TCP/IP to Network Manager of the same PC as the Network Manager
- System Setup and Configuration
 - Control calibration process
 - Setup software segmentation and zoning
 - Setup calibrated thresholding
 - Adjust processing parameters
 - Assign relays

UCM Software

- Maintenance and Diagnostic Tools
 - Real-time monitoring of sensor parameters
 - Plot mode
 - Review logged data
 - Pinpoint causes of nuisance alarms

Network Manager and StarNet

- A Windows®-Based Program Running on a PC
 - Communicates via NIU with the processors on the Silver Network
 - Sends maintenance and diagnostic data to UCM
 - Sends alarm data to StarNet
- StarNet Display and Control Software
 - Colour graphic display
 - Zone comprising one or more display lines
 - Star at target location on display line
 - Shooting star target trail

OMNITRAX

StarNet Display



OmniTrax vs Perimitrax®

- Pinpoint target location
- Multiple zones per cable
- 400 m (1,312 ft) vs. 200 m (656 ft)
- Enhanced data communications
- Easier to install
- Improved performance
 - Calibrated thresholds
 - Frequency diversity

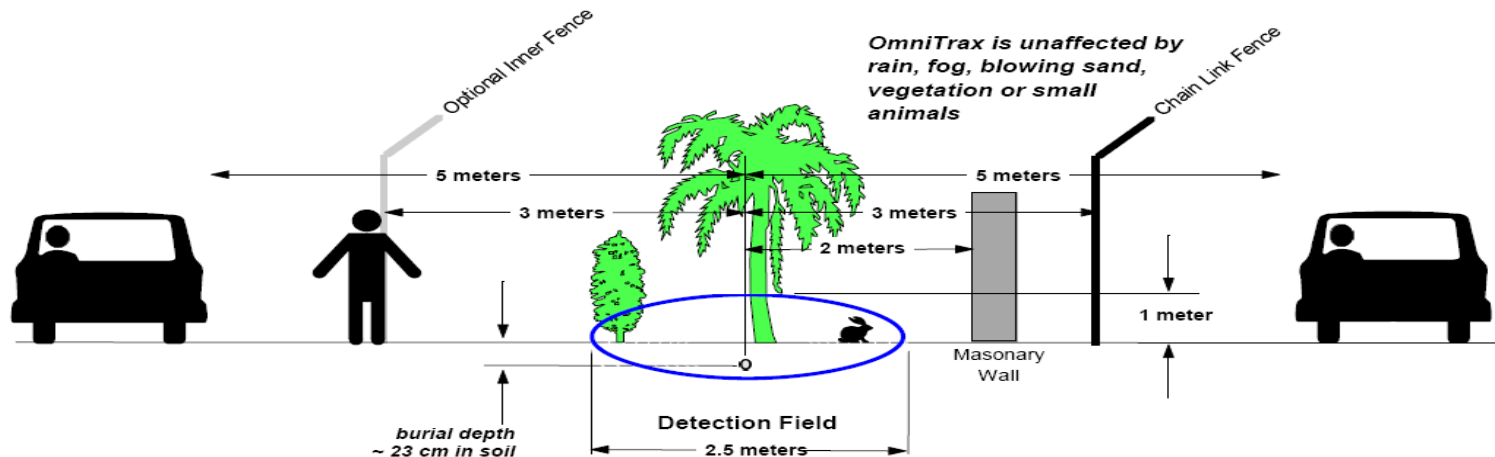
Application Considerations

- OmniTrax can be buried in virtually any type of material
 - Sand, clay, topsoil, rock, concrete, asphalt
 - Can have multiple mediums on a cable
 - Buried 23 cm (9 in) in soft materials, 6 cm (2.5 in) in hard materials
 - Relaxed installation criteria
- OmniTrax can be installed in most any type of terrain
 - Flat, hilly, around corners
 - Not seismically sensitive
- Surface may include lawns (including sprinkler systems), crushed stone, including landscaped areas with shrubs, flowers and trees
- Retrofit on Perimitrax and some Sentrax cables (contact factory)

Application Guidelines

- Sensor cables may cross large non-metallic water pipes by shielding area with sheet metal
- Sensor cables may be installed in areas with high voltage power lines (e.g. 400 kV) and where radar and communications devices are located (e.g. airports, military installations)
 - OmniTrax uses an ultra-wide band RF signal that will not interfere with communications
- In high lightning incidence areas, external gas discharge lightning arrestors are recommended to further protect electronics
- CCTV or other visual means of alarm assessment is highly recommended to confirm alarms and required level of response
- Consult Senstar for special site application questions

Clearances



Installation Considerations

1. Ensure proper clearance from all obstacles per above diagram.
2. Consult Site Survey Guide or Senstar-Stellar for clearance requirements from underground conduits & pipes.
3. Clear zone must have proper drainage to prevent water from collecting on top of sensor cable.
4. Can be installed in uneven terrain and around corners. Consult Site Survey Guide or Senstar-Stellar for minimum turning radius.
5. Cables are designed for direct burial - do not install in conduits.

OMNITRAX

Installation Through Trees



OMNITRAX

Installation Through Mixed Media

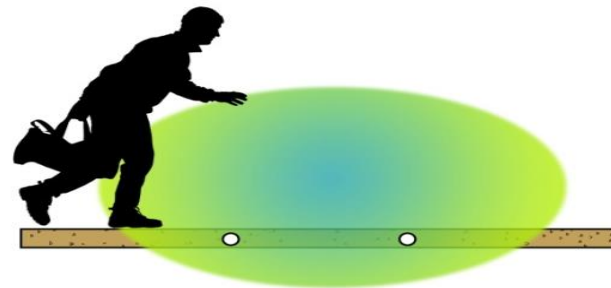


Rejection of Nuisance Alarms From Small Animals



Benefits of Ported Coax Technology

- HF / VHF Operation Provides
 - All-weather performance
 - Optimum detection of humans
 - Maximum rejection of small animals
 - Immune to vegetation and blowing debris
- Graded Cable
 - Uniform electromagnetic fields
 - Wide range of soil conductivity
 - Covert operation has lowest Vulnerability to defeat (Vd)



Ported coax is the only sensor technology classified as active, volumetric, terrain-following and covert

Summary

- Pinpoint Target Location
 - Star on map display
 - Software zoning
 - Improved CCTV assessment
- Improved Performance Through
 - Calibrated thresholding
 - Software segmentation
- Power and Data Options
 - Over sensor cables
 - Copper and fiber interface options

Thank You.